

An Exact Fifteen-Dimensional Local Koopman Jet for a Polynomial Hénon Germ

Anonymous Route-A report

Abstract

We construct the exact pullback of $F(u, v) = (u^2 + 3u/2 - v/2, u)$ on $A_4 = \mathbb{Q}[u, v]/(u, v)^5$. The maximal-ideal filtration proves a five-block factorization of its finite characteristic determinant. This is a local operator prefix, not a global Fredholm claim.

1 Well-defined pullback and convention

Put $\mathfrak{m} = (u, v)$. Since $F(0) = 0$, both coordinates of F lie in $\mathfrak{m}$, so pullback sends $\mathfrak{m}^5$ into itself. Therefore

$$K[p] = [p \circ F] : A_4 \longrightarrow A_4$$

is well defined. We order monomials by total degree and descending u power within each degree. Column j is the coefficient vector of K applied to the j th basis element.

2 Filtration proof

On the degree- d associated quotient, only

$$L(u, v) = (3u/2 - v/2, u)$$

survives. Its eigenvalues are 1 and $1/2$, so the homogeneous degree- d pullback has eigenvalues $1, 1/2, \dots, 2^{-d}$.

d	dimension	eigenvalues	trace	determinant
0	1	1	1	1
1	2	1, $1/2$	$3/2$	$1/2$
2	3	1, $1/2$, $1/4$	$7/4$	$1/8$
3	4	1, $1/2$, $1/4$, $1/8$	$15/8$	$1/64$
4	5	1, $1/2$, $1/4$, $1/8$, $1/16$	$31/16$	$1/1024$

Proposition 1. *The finite characteristic data are*

$$\chi_K(\lambda) = \prod_{k=0}^4 (\lambda - 2^{-k})^{5-k}, \quad \det(I - zK) = \prod_{k=0}^4 (1 - 2^{-k}z)^{5-k}.$$

Hence $\text{tr } K = 129/16$ and $\det K = 2^{-20}$.

Proof. The degree ordering makes K block triangular. Multiplying the five diagonal block polynomials gives the assertion. $\square$

3 Boundary

The result concerns a fifteen-dimensional quotient only. This revision does not yet audit the nonlinear correction against the linearized matrix or give the full hostile-replay provenance. No global Koopman spectrum, invariant Banach space, nuclearity, Fredholm determinant, arithmetic data, Hilbert–Pólya operator, or Route-B authorization is claimed. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.